Multimodal Sentiment Analysis Using Audio–Visual Fusion

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***Abstract*—Sentiment analysis has traditionally focused on textual data, which often overlooks important non-verbal cues such as facial expressions and prosody. This work presents a multimodal sentiment analysis system that integrates visual and textual information extracted from video and speech to achieve a more robust and nuanced understanding of affect. Faces are detected in video frames and passed to a facial expression classifier based on the POSTER architecture, while the audio stream is transcribed using a Whisper-based pipeline and analyzed with an LSTM text classifier. The two modality-specific predictions are fused through a softmax-based probability combination to obtain a final sentiment label. The system further translates the recognized text into Turkish and overlays subtitles together with the predicted sentiment for each utterance. Qualitative and quantitative results indicate that combining modalities leads to more stable predictions than relying on a single modality, especially in challenging conditions such as noisy audio or ambiguous facial expressions.**

***Index Terms*—multimodal sentiment analysis, facial expression recognition, speech analysis, LSTM, transformers, audio–visual fusion**

I. Introduction

Sentiment analysis, or opinion mining, aims to determine the affective state expressed in human communication. While early work has focused primarily on written text, real-world interactions are inherently multimodal: speakers communicate emotions through lexical choice, prosody, facial expressions, and other non-verbal cues. Systems that rely on a single modality may therefore miss important context, leading to brittle or biased predictions.

To address this limitation, multimodal sentiment analysis combines complementary information sources, typically text, audio, and vision, to form richer representations of affect. This is particularly important in scenarios such as social media video analysis, customer support interactions, and human–computer interaction, where subtle differences in tone or expression can significantly change the interpretation of a message.

In this work, we develop a prototype multimodal sentiment analysis pipeline that processes short video clips containing spoken utterances. The system integrates:

- a visual pathway based on face detection and a transformer-based facial expression classifier (POSTER),
- a speech pathway that transcribes the audio with a Whisper model and feeds the resulting text to an LSTM-based sentiment classifier,
- a fusion mechanism that combines the modality-specific sentiment probability distributions via softmax and elementwise product.

For each spoken segment, the system outputs a final sentiment label and overlays it with Turkish subtitles derived from the translated transcript.

The main contributions of this work are:

- a complete end-to-end pipeline for audio–visual sentiment analysis from raw video input,
- a simple yet effective probability-level fusion scheme to integrate visual and textual predictions,
- a practical demonstration of multilingual output through automatic translation and subtitle rendering.

II. Related Work

Multimodal affect recognition has attracted increasing attention with the availability of large-scale datasets and powerful deep architectures. Many approaches combine facial expression recognition, acoustic emotion recognition, and textual sentiment analysis to leverage complementary cues.

In the visual domain, transformer-based models have recently achieved strong performance on facial expression recognition benchmarks. POSTER [2] introduces a pyramid cross-fusion transformer network tailored for robust expression recognition under varied poses and occlusions. Toolboxes such as FaceX-Zoo [3] provide flexible implementations for face detection and recognition, enabling rapid prototyping of vision pipelines.

For video-based facial representation learning, masked autoencoder approaches such as MARLIN [1] demonstrate the benefits of self-supervised pretraining on large facial video corpora. On the language side, recurrent neural networks and variants of LSTMs remain popular for modeling sequential text in sentiment analysis tasks, especially when training data is moderate in size.

Our work draws on these developments by combining a transformer-based facial expression model with an LSTM text classifier operating on transcribed speech. Instead of designing

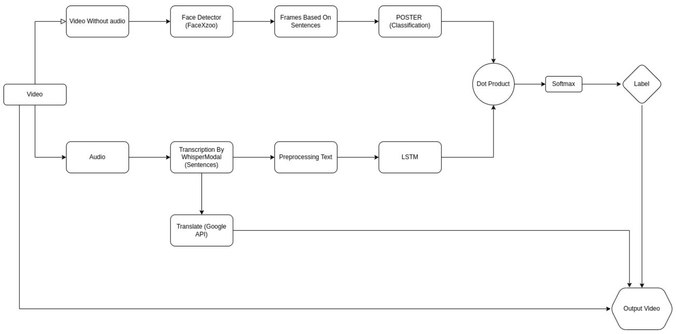

Fig. 1. Overall multimodal sentiment analysis pipeline integrating visual and textual pathways.

a complex cross-attention fusion module, we focus on a lightweight probability-level fusion strategy that is easy to implement and interpret in a practical system.

III. Proposed Method

A. System Overview

The input to the system is a video clip containing one or more spoken sentences. The pipeline consists of three main stages:

1) visual processing of selected frames aligned with spoken segments,
2) audio transcription and text-based sentiment analysis,
3) fusion of the modality-specific sentiment distributions and rendering of subtitles.

An overview of the processing flow is shown in Fig. 1. Starting from the raw video, we:

- extract the audio track and video frames,
- perform face detection and select representative frames for each utterance,
- classify facial expressions with POSTER,
- transcribe the audio with a Whisper-based model,
- preprocess the transcript and classify sentiment with an LSTM model,
- fuse the probability distributions and translate the transcript into Turkish,
- overlay subtitles and sentiment labels on the output video.

B. Visual Modality: Facial Expression Analysis

For the visual pathway, we first perform face detection on video frames using FaceX-Zoo [3] to isolate regions of interest. The video is segmented into utterance-level intervals, and for each segment we select one or more key frames. These frames are passed to the POSTER model [2], a pyramid cross-fusion transformer network designed for facial expression recognition.

POSTER is trained on the RAF-DB dataset, which contains facial images labeled with basic emotion categories such as *Surprise*, *Fear*, *Disgust*, *Happiness*, *Sadness*, *Anger*, and *Neutral*. For each frame, POSTER outputs a probability distribution over these classes. For segments with multiple frames, we average the logits or probabilities to obtain a single visual sentiment distribution per utterance.

C. Text Modality: Transcribed Speech Sentiment

The audio stream is transcribed into English text using a Whisper-based speech recognition model. Accurate transcription is crucial, as errors propagate into the downstream sentiment classifier. After transcription, we apply standard text preprocessing, including lowercasing, tokenization, optional stopword handling, and normalization of punctuation.

The cleaned text is then encoded and fed to an LSTM-based classifier. The LSTM processes the token sequence and produces a hidden representation that is mapped to the same set of emotion labels as the visual model (e.g., *Surprise*, *Fear*, *Disgust*, *Happiness*, *Sadness*, *Anger*, *Neutral*). The output layer uses a softmax activation to produce a probability distribution over emotion classes.

The text sentiment classifier is trained on a labeled corpus of short text segments such as tweets or sentence-level utterances, annotated with the same emotion categories. This alignment of label spaces enables straightforward fusion between the two modalities.

D. Multimodal Fusion

Let $p^{(v)} \in \mathbb{R}^C$ denote the softmax probability vector produced by the visual classifier for a given utterance, and $p^{(t)} \in \mathbb{R}^C$ the corresponding probability vector from the text classifier, where C is the number of emotion classes.

We first ensure that both vectors are normalized with softmax. To combine them, we compute an elementwise product:

$$p_c^{(\text{fusion})} \propto p_c^{(v)} \cdot p_c^{(t)}, \tag{1}$$

followed by a normalization step:

$$p_c^{(\text{fusion})} = \frac{p_c^{(v)} \cdot p_c^{(t)}}{\sum_{k=1}^{C} p_k^{(v)} \cdot p_k^{(t)}}. \tag{2}$$

This multiplicative fusion scheme naturally emphasizes classes where both modalities are confident and downweights classes with disagreement. The final sentiment label is obtained as:

$$\hat{y} = \arg\max_c p_c^{(\text{fusion})}. \tag{3}$$

E. Translation and Subtitle Rendering

To make the system accessible to Turkish-speaking users, we translate the English transcript of each utterance into Turkish using the `googletrans` library, which acts as a Python interface to the Google Translate API. For each segment, we display the Turkish subtitle together with the fused sentiment label (e.g., "Happiness") over the corresponding portion of the video. This results in an intuitive visualization where viewers can see both the translated content and the model's affective interpretation of the speaker.

IV. Experimental Setup

A. Data

We evaluate the system on a collection of short video clips containing spoken sentences. Each video is segmented into utterances aligned with the audio, and one or more representative frames are selected per utterance. The corresponding

	RAF-DB	
Components	Acc	Acc(mean)
(a) Landmark only	80.08	72.21
(b) Image only	90.51	82.73
(c) Baseline	91.00	84.64
(d) Baseline+pyramid	91.27	85.66
(e) Baseline+cross_fusion	91.63	85.01
(f) POSTER	**92.05**	**86.03**

Fig. 2. Component analysis of the POSTER facial expression classifier on RAF-DB.

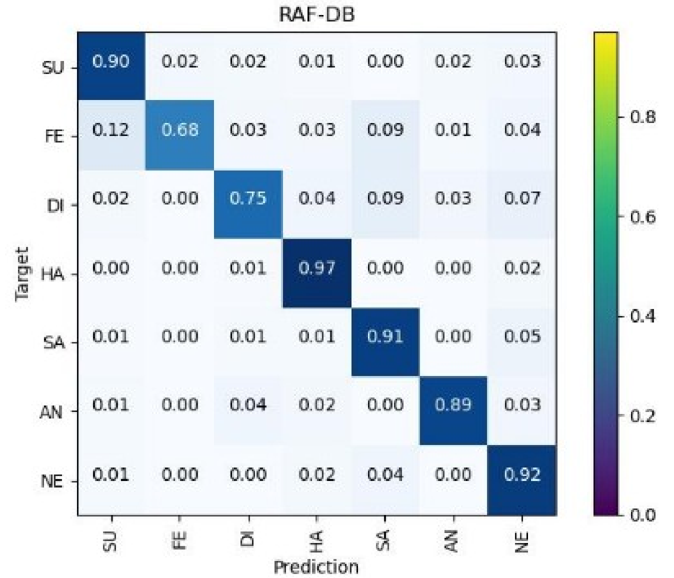

Fig. 3. Confusion matrix of the POSTER model on RAF-DB.

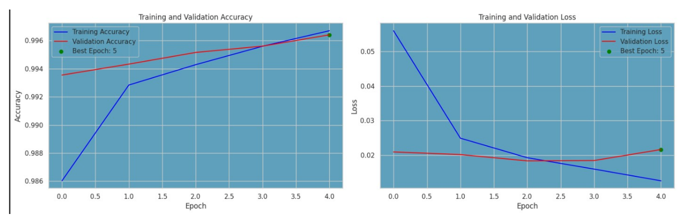

Fig. 4. Training and validation accuracy (left) and loss (right) for the LSTM text classifier.

transcripts and translated subtitles are generated automatically, and the emotion labels predicted by each modality are compared qualitatively across examples.

B. Implementation Details

Face detection and alignment are implemented using the FaceX-Zoo toolbox [3]. The POSTER model [2] is used as a pretrained backbone for visual sentiment classification. For text, a Whisper-based model is employed for speech-to-text conversion, and the LSTM classifier is implemented in PyTorch with a standard architecture (embedding layer, single or multi-layer LSTM, and a linear classification head).

During inference, the visual and text classifiers operate independently, and their probability outputs are fused as described in Section III-D. For subtitle rendering and translation, we use the `googletrans` Python package.

V. Results and Discussion

A. Visual Modality Performance

Fig. 2 summarizes the component analysis of the POSTER architecture on the RAF-DB dataset, comparing different design variants. The full POSTER model achieves the highest average accuracy, demonstrating the benefit of pyramid cross-fusion for robust facial expression recognition.

The confusion matrix in Fig. 3 illustrates the distribution of predictions across emotion classes. As expected, expressions such as *Happiness* and *Neutral* tend to be recognized more reliably, while more subtle or ambiguous emotions can be harder to distinguish. Visually similar emotions such as *Fear* and *Surprise* are a common source of confusion, which aligns with prior work on facial expression recognition.

B. Text Modality Performance

The LSTM-based text classifier exhibits strong performance on the training corpus. Fig. 4 shows the training and validation accuracy and loss across epochs. Both metrics indicate stable convergence without severe overfitting, with validation accuracy closely tracking training accuracy.

The confusion matrix in Fig. 5 reveals that most errors occur between neighboring or conceptually related emotion classes, while the majority class is predicted with very high accuracy. The detailed classification report in Fig. 6 confirms that precision, recall, and F1-score are all high across classes, with macro and weighted averages close to 0.98 or above.

In practice, transcription errors and disfluencies can degrade text-based performance. Nonetheless, the LSTM classifier generally produces stable probability distributions over the emotion classes, which are suitable for fusion with the visual model.

C. Effect of Multimodal Fusion

The multiplicative fusion scheme described in Section III-D yields more robust predictions than either modality alone. In qualitative analysis of multiple clips, we observe three typical behaviors:

- When both modalities agree with high confidence, fusion reinforces the shared class and yields a very sharp distribution.
- When one modality is uncertain (e.g., nearly uniform probabilities), fusion allows the more informative modality to dominate.
- When modalities disagree, the product tends to favor classes where neither probability is extremely low, effectively downweighting unlikely interpretations.

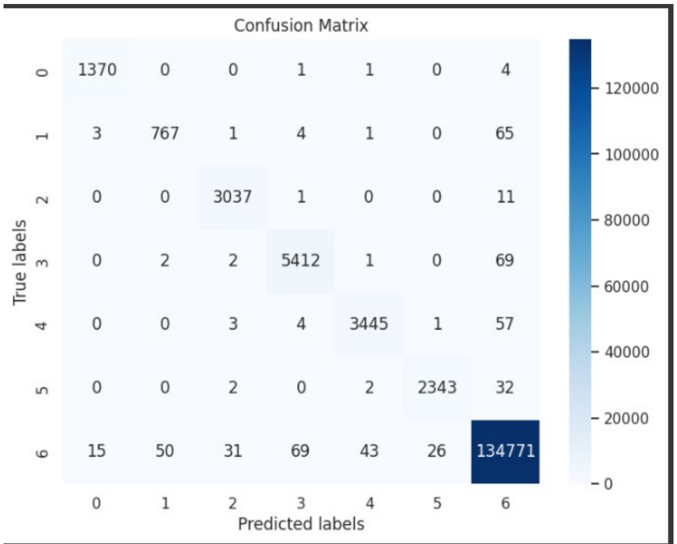

Fig. 5. Confusion matrix of the LSTM text classifier.

	precision	recall	f1-score	support
0	0.99	1.00	0.99	1376
1	0.94	0.91	0.92	841
2	0.99	1.00	0.99	3049
3	0.99	0.99	0.99	5486
4	0.99	0.98	0.98	3510
5	0.99	0.98	0.99	2379
6	1.00	1.00	1.00	135005
accuracy			1.00	151646
macro avg	0.98	0.98	0.98	151646
weighted avg	1.00	1.00	1.00	151646

Fig. 6. Classification report of the LSTM text classifier showing precision, recall, and F1-scores per class.

This behavior is particularly beneficial in noisy conditions, such as low lighting (affecting the visual modality) or background noise (affecting the speech modality).

VI. Conclusion and Future Work

We presented a multimodal sentiment analysis system that combines facial expression recognition and transcribed speech sentiment analysis to infer emotions in short video clips. The pipeline integrates existing components—FaceX-Zoo for face detection, POSTER for visual sentiment classification, a Whisper-based transcription model, and an LSTM text classifier—and fuses their outputs through a simple probability-level fusion strategy. The system additionally supports automatic translation into Turkish and subtitle overlay, enabling an accessible user-facing visualization of both content and sentiment.

Future work includes extending the system to incorporate acoustic features directly, exploring more advanced cross-modal fusion mechanisms (e.g., transformers with cross-attention), and scaling evaluation to larger, publicly available multimodal datasets with ground-truth emotion labels. Another promising direction is to investigate domain adaptation techniques to improve robustness across different recording conditions and cultural contexts.

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